CS-26-009פרויקט מס׳

Data-Driven Risk-Adjusted Performance Analysis

ניתוח ביצועים מותאמי-סיכון מבוסס נתונים

פרויקט מחקרי

דוח מסכם

הוכן לשם השלמת הדרישות לקבלת

תואר ראשון במדעי הטבע B.Sc.

מאת

נהוראי שדה ואביב אסרף

בהנחיית ד״ר דביר רוס

הוגש למחלקה למדעי המחשב

המכללה האקדמית להנדסה סמי שמעון

באר-שבע

תאריך עברי: ט״ז סיון התשפ״ו תאריך לועזי: 02/06/2026

המכללה האקדמית להנדסה ע"ש סמי שמעון

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פרויקט מחקרי — דוח מסכם

מאת נהוראי שדה ואביב אסרף

חתימת סטודנט 1 : ____________________ תאריך : ________________

חתימת סטודנט 2 : ____________________ תאריך : ________________

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אישור ועדת הפרויקטים : ________________ תאריך : ________________

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תקציר

פרויקט זה בוחן לעומק את מדדי הביצוע מותאמי-הסיכון הנפוצים — יחסי Sharpe, Sortino, Calmar, Treynor ו-Information — ומציע מדד משלים חדש המבוסס על יציבות הצמיחה האקספוננציאלית של מחירי נכסים. המדד החדש, $diff \times r \times b$, נגזר מרגרסיה אקספוננציאלית על המחיר ההיסטורי: diff הוא המרחק בין קו המגמה למחיר הנוכחי, r הוא מקדם המתאם (טיב ההתאמה למגמה), ו-b הוא שיפוע הצמיחה היומי.

בעבודה אמפירית רחבה על 100 קרנות סל (ETFs) ובמשך תשע תקופות שוק שונות, מצאנו כי המדד החדש מתפקד כסיגנל של חזרה-לטרנד (mean-reversion), בעוד שהיחסים הקלאסיים מתנהגים בפועל כמו סיגנלי מומנטום. כתוצאה מכך, המדד החדש מנצח באופן מובהק בנקודות היפוך ובשווקים תנודתיים (למשל, דיוק כיווני של 97% מול 9% בהתאוששות ממשבר הקורונה), אך נחות במגמות מתמשכות. אנו מסיקים כי שני סוגי המדדים משלימים ולא מתחרים, וכי שילוב ביניהם לפי משטר השוק עדיף על כל אחד מהם בנפרד.

Abstract

Risk-adjusted performance measures—the Sharpe, Sortino, Treynor, Calmar, and Information ratios—are central to investment evaluation, yet they rely on simplifying assumptions that can mislead under shifting market regimes. This project studies these classical ratios and proposes a complementary structural measure built on the stability of exponential compounding. The proposed score, diff × r × b, is derived from an exponential regression of price: diff = ExpReg(last) − Price(last) is the distance of price from its fitted trend, r is the Pearson correlation of log-price with time (trend-fit quality), and b is the daily log-growth rate.

We evaluate the score empirically against the five classical ratios on a universe of 100 equity ETFs across nine market eras and a 180-configuration window sweep, measuring both directional accuracy and the realized return of a long/short portfolio. The central finding is that diff × r × b behaves as a mean-reversion-to-trend signal, whereas the classical ratios behave as momentum signals. Consequently the new measure decisively outperforms at reversals and in choppy markets (e.g., 97% vs. 9% directional accuracy on the post-COVID-crash rebound) and underperforms in sustained trends. We conclude that the two signal families are complementary rather than competing, and that a regime-aware combination dominates either one alone.

1. Introduction

Risk-adjusted performance measures play a central role in financial decision-making. By combining return with a measure of risk—total volatility, downside volatility, systematic risk, or drawdown—they allow investors to compare assets whose raw returns differ substantially. These ratios sit at the heart of portfolio construction, risk management, and the performance evaluation of both individual assets and complete strategies.

Despite their ubiquity, the classical ratios rest on simplifying assumptions that do not always reflect real markets. Volatility is treated as a symmetric proxy for risk even though investors experience gains and losses asymmetrically; return distributions are fat-tailed; and, crucially, market regimes shift. A measure that ranks an asset highly in a calm uptrend may rank it very differently across a crash and the recovery that follows. As our literature review established, this regime dependence is well documented, and it motivates the search for complementary information that the classical ratios do not capture.

Our progress report introduced such a candidate: a structural measure based on how closely an asset's price follows a smooth exponential compounding path. The present report carries that idea from concept to empirical test. We formalize the exponential-fit score diff × r × b, embed it in a directional-prediction and portfolio-backtest framework alongside the five classical ratios, and run a systematic experiment whose explicit purpose is to answer a single question: in which market conditions does the new measure add value relative to the classical ratios, and in which does it fail? Identifying these regimes—rather than claiming a universally superior ratio—is the contribution of this work.

2. Background

Risk-adjusted performance measures evaluate investment outcomes by balancing return against risk. The literature has proposed many such ratios, each emphasising a different facet of risk; their multiplicity reflects the complexity of financial risk rather than a deficiency of any one metric. The five measures studied in this project are defined below, with Rp the asset return, Rf the risk-free rate, Rb the benchmark return, σp total volatility, σd downside deviation, βp the beta to the benchmark, and Dmax the maximum drawdown.

- **Sharpe ratio (Sharpe, 1966):** (Rp − Rf) / σp — excess return per unit of total volatility.

- **Sortino ratio (Sortino & Price, 1994):** (Rp − Rf) / σd — excess return per unit of downside volatility only.

- **Treynor ratio (Treynor, 1965):** (Rp − Rf) / βp — excess return per unit of systematic (market) risk.

- **Information ratio (Grinold, 1989):** (Rp − Rb) / σ(Rp − Rb) — consistency of out-performance over a benchmark.

- **Calmar ratio (Young, 1991):** (Rp − Rf) / Dmax — annual return relative to the worst peak-to-trough loss.

Each ratio embeds specific assumptions and is suited to particular conditions; different ratios can therefore rank the same asset differently, especially across market regimes. The objective of this project is to examine these strengths and weaknesses systematically and to assess whether a complementary measure based on exponential-compounding stability can address some of their limitations by focusing on the stability of the return-generating process rather than on a single risk proxy.

3. Related Work

Improvements and alternatives to traditional ratios

Recognising that a single volatility number cannot capture the full complexity of risk, researchers have proposed measures such as the Omega ratio, which considers the entire return distribution, and the Upside Potential ratio, which weighs the frequency and magnitude of returns above a target. The consensus these efforts express—that volatility alone is incomplete and additional structural information is needed—directly motivates our study of where the classical ratios succeed or fail.

Compounding effects and exponential price behavior

Because asset prices evolve multiplicatively rather than additively, prices are commonly modelled with log-returns and log-linear regressions. Stable compounding appears as a near-linear curve in log-scale, while instability, regime shifts, or inconsistent return patterns weaken the exponential fit. This provides the theoretical foundation for our measure, which evaluates how consistently an asset follows an exponential compounding structure rather than merely whether it trends upward.

Behavior of ratios under trending markets

Empirical evidence shows that risk-adjusted measures are regime-dependent. Jegadeesh and Titman (1993) demonstrate strong momentum effects in sustained trends, which can lead volatility-based ratios such as Sharpe and Sortino to overstate performance during prolonged growth. Moskowitz, Ooi, and Pedersen (2012) show that time-series momentum performs well in stable trends but suffers sharp reversals at abrupt transitions, and Fama and French (2012) document substantial variation of momentum effects across conditions. These findings imply that classical risk-adjusted ratios react implicitly to trend structure even when trend is not modelled explicitly—an observation our experiment confirms directly.

Regression-based analysis of price structure

Linear and log-linear regressions are widely used to estimate growth rates and assess the stability of return-generating processes (Campbell, Lo & MacKinlay, 1997). A strong fit in log-price space indicates stable compounding, whereas a weak fit signals noise, regime shifts, or structural breaks (Tsay, 2010); residual diagnostics help detect deviations from expected growth (Brockwell & Davis, 2016). Our exponential-fit measure follows this tradition, using the regression not to forecast prices but to quantify the consistency of the compounding process itself.

Limitations of classical ratios

The literature documents several structural limitations of classical ratios: volatility clustering violates the constant-variance assumption underlying Sharpe and Sortino (Engle, 1982); fat tails and excess kurtosis cause volatility-based measures to understate downside risk (Cont, 2001); drawdown measures such as Calmar capture peak-to-trough loss but miss intermediate instability (Magdon-Ismail et al., 2004); and rankings differ materially across bull, bear, and crisis regimes (Kritzman & Li, 2010; Daniel & Moskowitz, 2016). Together these justify exploring whether structural information—the stability of exponential compounding—can complement classical ratios under non-stationary conditions.

4. Method

Classical risk-adjusted ratios

We compute all five classical ratios on the training window from daily simple returns, with a risk-free rate of 4.5% per year and annualisation to 252 trading days. Sharpe and Sortino use total and downside volatility respectively; Treynor uses the beta of the asset's returns to the benchmark; the Information ratio uses the mean and standard deviation of the active return (asset minus benchmark); and Calmar divides the annualised excess return by the maximum drawdown over the window. A ratio that cannot be defined (for example, Sortino when there is no downside volatility) is treated as missing for that asset.

The exponential-fit residual and the diff × r × b score

Following the structural-stability idea from our literature review, we fit an exponential model y = a · e^(b ·t) to the training-window close price by performing an ordinary least-squares regression of log(price) on time t. From this fit we extract three quantities:

- **r** — the Pearson correlation between log(price) and time, a measure of how cleanly the asset follows a smooth exponential path (trend-fit quality, $|r| \leq 1$);
- **b** — the slope, i.e. the daily log-growth rate (the long-run trend direction and strength);
- **diff** = ExpReg(last) − Price(last) — the signed distance between the fitted trend and the actual price on the last training day.

The composite score is their product, Score = diff × r × b. Its sign has a direct interpretation. When the price sits below its own smooth growth path (diff > 0) on a well-fit uptrend (r > 0, b > 0), the score is positive: a structurally stable, upward-trending asset currently trading below its trend. When the price has overshot its trend (diff < 0), the score is negative. The score is therefore a mean-reversion-to-trend signal—it expects price to return toward its exponential path—modulated by trend quality (r) and slope (b). Figure 1 illustrates both signs on two well-fit ETFs.

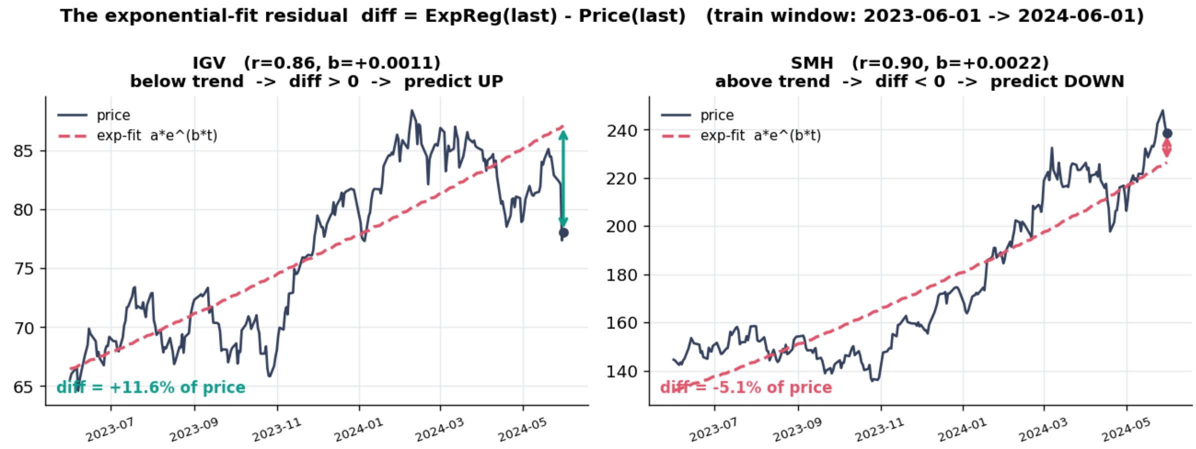

Figure 1. *The exponential-fit residual diff for two well-fit uptrending ETFs. Left: price below its fitted trend (diff > 0) → the score predicts a move up. Right: price above its trend (diff < 0) → the score predicts a move down.*

From score to directional prediction

To compare measures on equal footing we reduce each to a directional prediction. For a given measure, the predicted direction for an asset is the sign of its score over the training window: a positive score predicts the price will rise over the subsequent test window (UP), a negative score predicts a fall (DOWN), and a zero or undefined score is non-decisive (FLAT). The realized direction is the sign of the actual test-window return. A prediction is correct when the predicted and realized directions match and the prediction is decisive. The directional accuracy of a measure is the fraction of decisive predictions that are correct, computed over the entire ETF universe; 50% is the coin-flip baseline.

Applying the same sign-of-score rule to every measure is what makes the comparison fair: the five classical ratios are positive for assets with strong recent risk-adjusted returns, so they implicitly predict continuation (momentum); the exponential-fit score is positive for assets trading below a stable trend, so it predicts reversion. The experiment measures which behavior pays off in which regime.

Long/short portfolio backtest

Directional accuracy treats every asset equally; to weight by conviction and obtain an economic measure we also build a dollar-neutral long/short portfolio for each measure. Assets are ranked by score; the top ten form the long basket and the bottom ten the short basket. Within a basket, weights are proportional to |score| (so higher-conviction names receive more capital), with $1,000 deployed per side. A strict short gate requires $r < 0$ and $b < 0$ for short eligibility, so that only structurally weak names are shorted. Each position's profit-and-loss is evaluated against its realized test-window return (longs profit when the asset rises, shorts when it falls), and the portfolio's total return is the net P&L divided by the capital deployed. This mirrors exactly the portfolio logic implemented in our Streamlit application.

5. Data

All market data are daily adjusted close prices retrieved from Yahoo Finance (via the yfinance library) with automatic adjustment for dividends and splits. The asset universe is a curated set of 100 equity exchange-traded funds (ETFs) spanning broad US indices, style and size factors, dividend and sector funds, thematic and innovation funds, and international regions and single countries. ETFs are used deliberately: as diversified baskets they suppress single-name idiosyncrasies and let the experiment isolate the regime behavior of the measures rather than company-specific noise. The benchmark for Treynor and the Information ratio is the S&P 500 index (^GSPC).

For every configuration the data are split into two consecutive windows: a training window on which all measures are computed, and an immediately following test window on which the realized direction and returns are observed. An asset is included only if it has at least ten training observations and five test observations and the exponential fit is defined; depending on the era this yields 98–100 usable ETFs (newer thematic funds lack history in the earliest eras). To make the sweep efficient and fully reproducible, the entire universe was downloaded once over 2017–2026 and cached locally; every configuration is then produced by slicing this cache, so identical inputs yield identical results.

6. Experimental Setup

The experiment is a systematic sweep designed to expose regime dependence. We vary three controls that together define a market scenario: the test-end date (which selects the market era), the training-window length, and the test-window length. We evaluate nine market eras chosen to span distinct regimes—from the COVID-19 crash and its rebound, through the 2021 momentum top and the 2022 bear market, to the 2024 bull market and the most recent data—crossed with five training windows (1, 2, 3, 6, and 12 months) and four test windows (1, 2, 3, and 6 months). This yields $9 \times 5 \times 4 = 180$ configurations. For each configuration we record, for all six measures, the directional accuracy and the long/short portfolio return defined in the Method section.

To summarise a measure's edge we compare the new score against the classical family: the accuracy advantage is the new score's accuracy minus the mean accuracy of the five classical ratios. When aggregating per era we retain only configurations in which the new score makes at least 30 decisive predictions, ensuring the reported accuracies rest on a meaningful sample. The nine eras and their aggregate results are listed in Table 1; four configurations, two where the new score wins and two where it loses, are then examined in detail as case studies.

Table 1. *Mean directional accuracy by market era, averaged over all 20 train/test window combinations per era (new score vs. the average of the five classical ratios). Advantage is in percentage points.*

Market era	**New (diff×r×b)**	**Classical (avg)**	**Advantage**
COVID crash	74.2%	25.5%	**+48.7**
2022 bear / chop	61.9%	38.6%	**+23.3**
early-2025	50.8%	43.6%	**+7.3**
2024 bull	57.7%	55.1%	**+2.6**
2021 momentum top	60.8%	58.3%	**+2.4**
recent (2026)	49.6%	58.6%	**-9.1**
2023 AI recovery	43.6%	55.5%	**-12.0**
post-COVID rally	53.9%	69.4%	**-15.5**
2022 bear decline	26.0%	69.0%	**-43.0**

7. Results and Future work

Global view: a regime-dependent edge

Averaged over every window combination, the new score's advantage over the classical ratios is strongly regime-dependent (Figure 2, Figure 3). It is largest in the COVID crash (+48.7 points of accuracy) and the choppy 2022 bear tape (+23.3), roughly neutral in trending bull phases, and most negative in the sustained 2022 decline (−43.0) and the post-COVID momentum rally (−15.5). The ordering is not random: the new score wins precisely where price action reverses or oscillates, and loses precisely where a single direction persists. This is the signature of a mean-reversion signal competing against momentum.

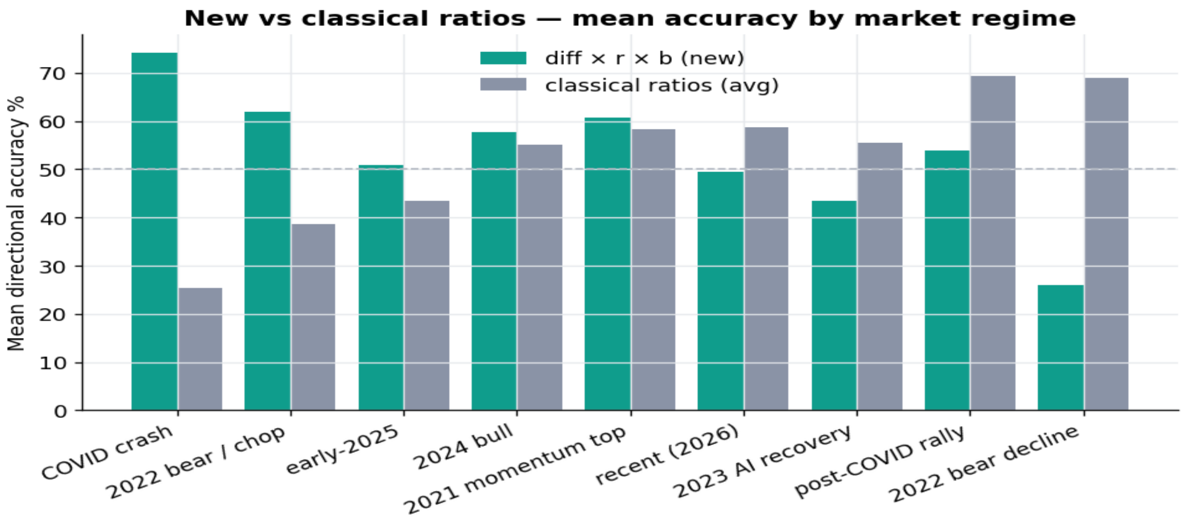

Figure 2. *Mean directional accuracy of the new score (teal) versus the average of the five classical ratios (grey) for each market era. The dashed line marks the 50% coin-flip baseline.*

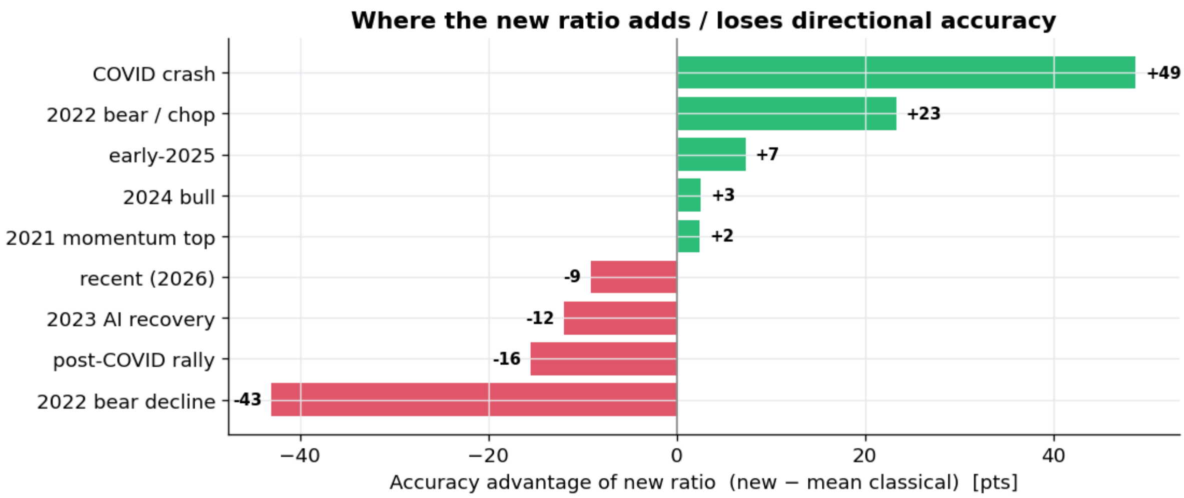

Figure 3. *Accuracy advantage of the new score (new minus mean-classical, in points) by era. Green bars are eras where the new score helps; red bars are eras where it hurts.*

Case study A — COVID crash rebound (new score wins)

Test-end 2020-04-15, with a 3-month training window (2019-12-15 → 2020-03-15) and a 1-month test window (→ 2020-04-15), over 98 ETFs. The training window ends near the 23 March 2020 bottom, so after the crash almost every ETF sat far below its exponential trend, giving a large positive diff and an UP prediction from the new score—exactly the violent rebound that followed. The classical ratios, depressed by catastrophic recent risk-adjusted returns, predicted continued declines. The result is the starkest in the study: 96.9% directional accuracy for the new score against a 9.2% average for the classical ratios, and a long/short return of +10.9% versus a −8.8% average (Table 2, Figure 4). The scatter in Figure 5 shows almost every name in the upper-right and upper-left quadrants where the new score's sign matches the realized move.

Table 2. *COVID crash rebound — per-formula directional accuracy and long/short portfolio return (98 ETFs).*

Formula	Directional accuracy	Decisive	L/S portfolio return
diff x r x b	**96.9%**	**98**	**+10.9%**
Sharpe	3.1%	98	-11.6%
Sortino	3.1%	98	-9.8%
Calmar	3.1%	98	-16.5%
Treynor	3.1%	98	-11.6%
Information	33.7%	98	+5.4%

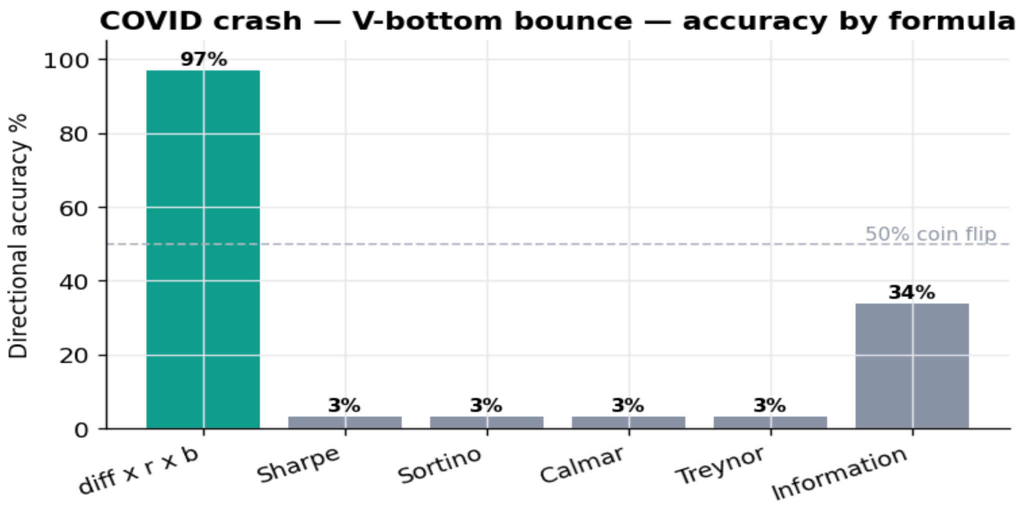

Figure 4. *COVID crash rebound: directional accuracy by formula. The new score (teal) reaches 96.9% while the classical ratios sit near 3%.*

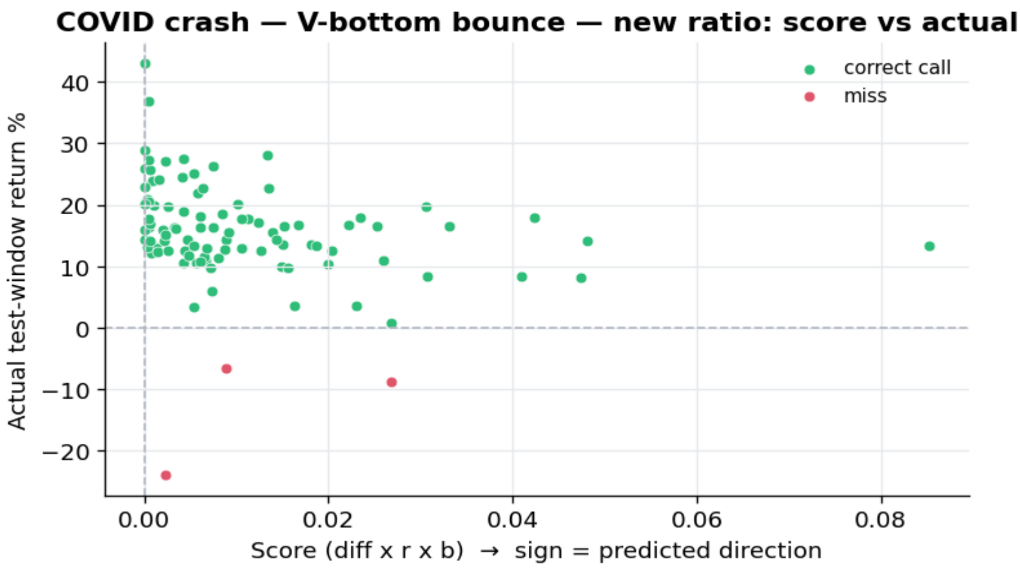

Figure 5. *COVID crash rebound: new-score value (x, sign = predicted direction) versus actual test-window return (y). Green = correct call, red = miss.*

Case study B — 2022 choppy bear tape (new score wins)

Test-end 2022-12-30, with a 12-month training window (2021-09-30 → 2022-09-30) and a 3-month test window, over 100 ETFs. The test window is the range-bound, mean-reverting Q4-2022 tape. Here too the reversion signal dominates: names stretched below their trend snapped back and names stretched above faded, giving the new score 90.0% accuracy against a 22.6% classical average, with a +8.1% long/short return versus +2.4% (Table 3, Figure 6). This case shows that the new score's edge is not unique to a single crash but generalises to choppy, directionless markets.

Table 3. *2022 choppy bear tape — per-formula accuracy and long/short return (100 ETFs).*

Formula	Directional accuracy	Decisive	L/S portfolio return
diff x r x b	**90.0%**	**100**	**+8.1%**
Sharpe	17.0%	100	+2.4%
Sortino	17.0%	100	+2.9%
Calmar	15.0%	100	+5.3%
Treynor	17.0%	100	+1.2%
Information	47.0%	100	+0.3%

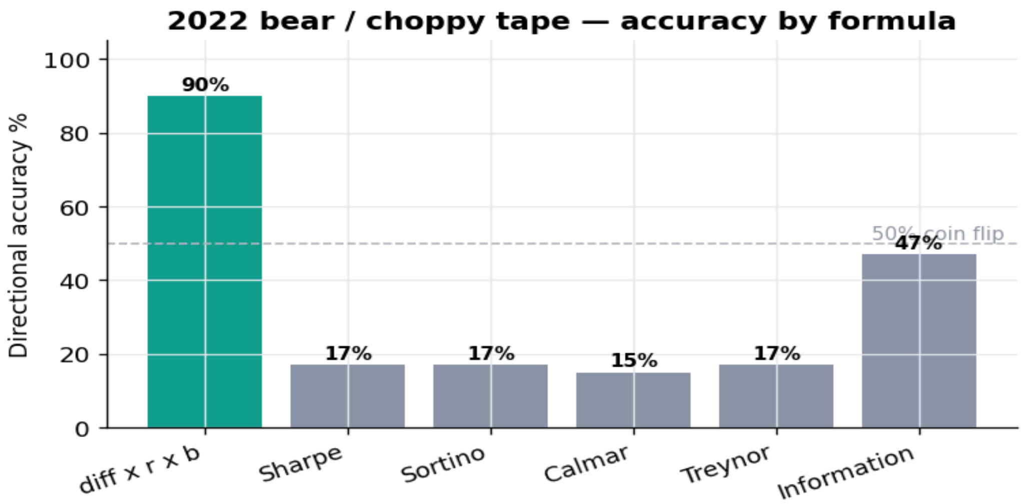

Figure 6. *2022 choppy bear tape: directional accuracy by formula. The new score (teal) reaches 90% against a low-20s classical average.*

Case study C — post-COVID momentum rally (new score loses)

Test-end 2020-12-31, with a 3-month training window (2020-08-30 → 2020-11-30) and a 1-month test window, over 99 ETFs. By late 2020 the market had been rallying for months and most ETFs traded above their trend, so the new score (diff < 0) predicted a pullback that never arrived. Momentum was the correct call, and the classical ratios captured it: they averaged 93.9% accuracy while the new score collapsed to 2.0% (Table 4, Figure 7). This is the mirror image of Case A and confirms the mechanism—when a trend persists, predicting reversion is systematically wrong.

Table 4. *Post-COVID momentum rally — per-formula accuracy and long/short return (99 ETFs).*

Formula	Directional accuracy	Decisive	L/S portfolio return
diff x r x b	**2.0%**	**99**	**+0.6%**
Sharpe	96.0%	99	+0.7%
Sortino	96.0%	99	+1.0%
Calmar	96.0%	99	+1.0%
Treynor	96.0%	99	+0.7%
Information	85.9%	99	+0.0%

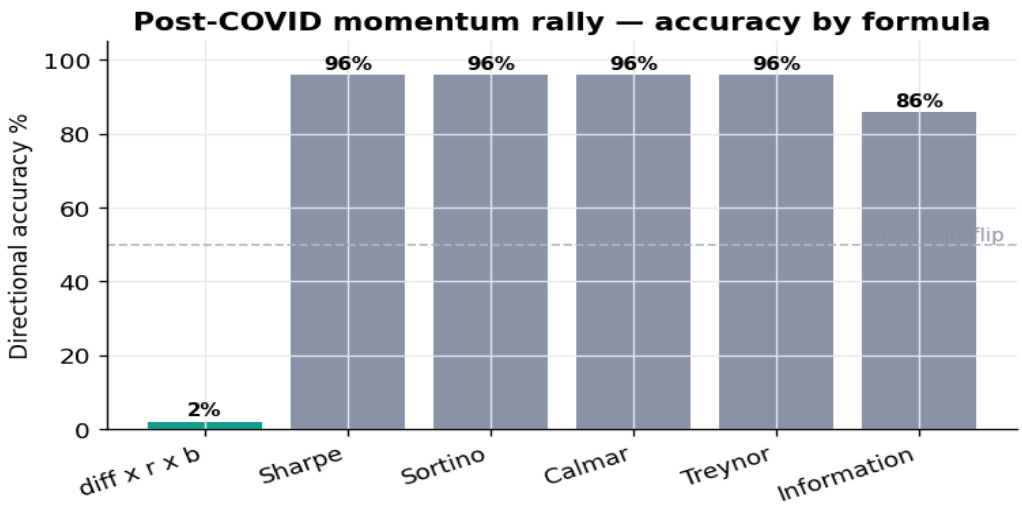

Figure 7. *Post-COVID momentum rally: directional accuracy by formula. The classical ratios (grey) dominate near 96% while the new score (teal) falls to ~2%.*

Case study D — 2022 sustained decline (new score loses)

Test-end 2022-06-15, with a 6-month training window (2021-09-15 → 2022-03-15) and a 3-month test window, over 99 ETFs. The test window is a sustained, directional sell-off. As the market kept falling, names below their trend kept falling further; the reversion signal repeatedly tried to catch a falling knife and was wrong. The classical ratios—deeply negative and therefore positioned short—earned 79.0% average accuracy and a +7.8% long/short return, against 8.1% accuracy and −6.4% for the new score (Table 5, Figure 8). The scatter (Figure 9) shows the failure plainly: most names carry a positive new-score (predict UP) yet realized sharply negative returns.

Table 5. *2022 sustained decline — per-formula accuracy and long/short return (99 ETFs).*

Formula	Directional accuracy	Decisive	L/S portfolio return
diff x r x b	**8.1%**	**99**	**-6.4%**
Sharpe	83.8%	99	+8.3%
Sortino	83.8%	99	+8.7%
Calmar	85.9%	99	+8.2%
Treynor	83.8%	99	+6.1%
Information	57.6%	99	+7.6%

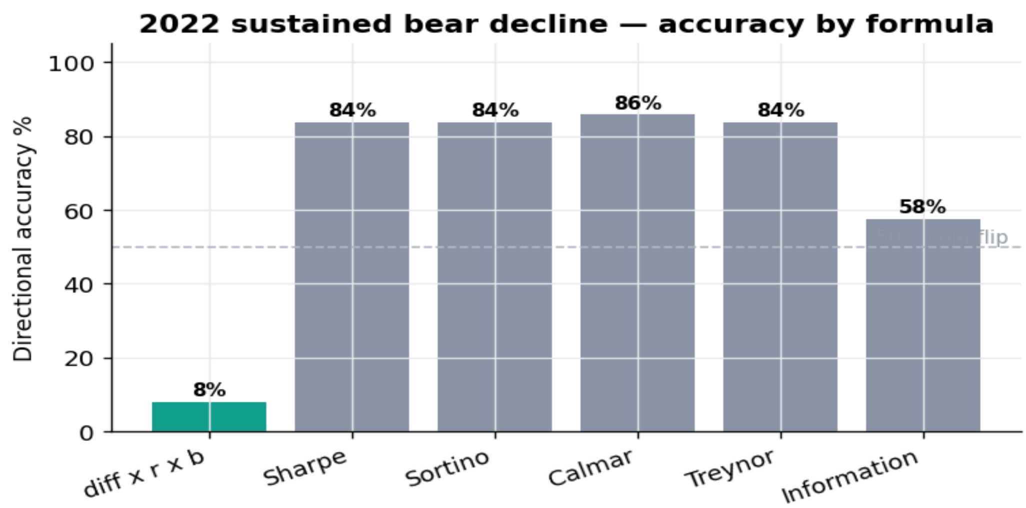

Figure 8. *2022 sustained decline: directional accuracy by formula. Momentum (classical, grey) wins; the reversion signal (teal) fails.*

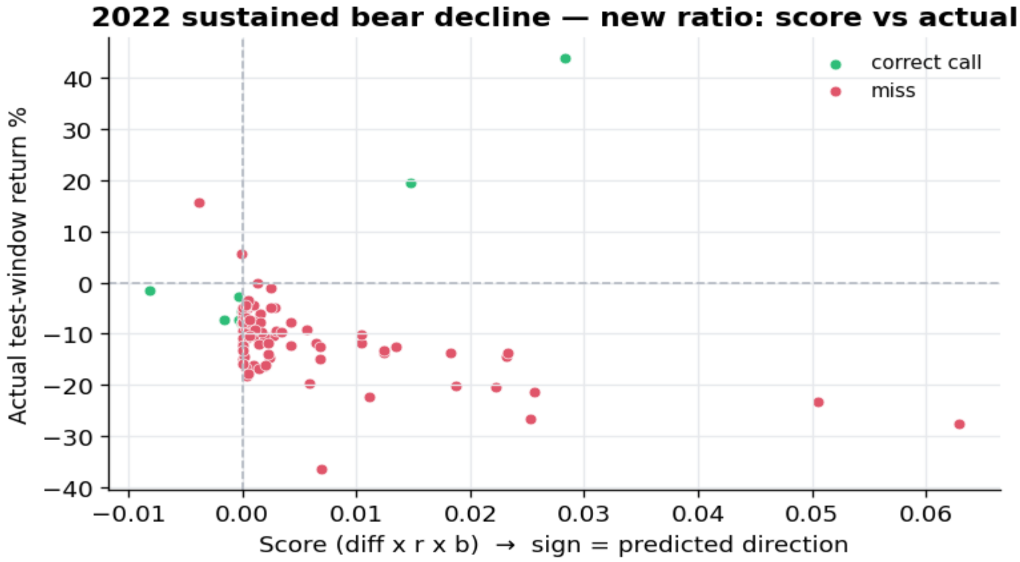

Figure 9. *2022 sustained decline: new-score value versus actual return. Most names have positive scores (predicting UP) yet fell — the falling-knife failure mode (red).*

Sensitivity to window length

The new score's edge is sharpest at short test windows immediately after a dislocation. In the COVID era a 1-month test captures the clean rebound, but as the test window lengthens to two or three months the rebound itself partially reverses, and the long/short portfolio return for the new score turns negative even though directional accuracy stays high. The training-window length matters less for direction than the era does, but very short training windows make r unreliable by capturing local noise, consistent with the preliminary observation in our progress report that windows shorter than about three months can be misleading. In practice the measure is best read over training windows of three months or more, with the understanding that its predictive horizon is short.

Discussion

The experiment gives a single, coherent explanation for every result. The exponential-fit score diff × r × b is a mean-reversion-to-trend signal: it is positive when price is below a stable trend and negative when above. The five classical ratios, by construction, are large after strong recent risk-adjusted returns and therefore act as momentum signals. The two families are thus near-opposite in sign exactly when the market is stretched, which is why their accuracies are almost mirror images across regimes (Table 1).

This connects directly to the related work in our literature review. Jegadeesh and Titman (1993) and Moskowitz, Ooi and Pedersen (2012) show that momentum is profitable within stable trends but suffers sharp reversals at transitions, and Daniel and Moskowitz (2016) document exactly such 'momentum crashes' in post-bear rebounds. Our Case A is a textbook momentum crash: the classical ratios, behaving as momentum, fail precisely where Daniel and Moskowitz predict they should, and the reversion-based new score succeeds there. The project therefore not only proposes a new measure but also provides direct empirical evidence for the implicit momentum sensitivity of classical risk-adjusted ratios that the literature had only inferred.

The practical implication is that the new score carries complementary, not redundant, information—precisely the working hypothesis stated in our progress report. Neither family is universally better; their value is conditional on the regime. A simple regime filter—using realized volatility, trend strength r, or the magnitude of diff itself to detect dislocations—could switch between reversion (diff × r × b) after dislocations and momentum (the classical ratios) inside established trends, and would be expected to outperform either family alone.

Future Work

The empirical evidence presented here rests on a static window sweep. For each configuration the training window is fixed, the measures are computed once, and a single test window is evaluated. This design is sufficient for the present question (in which regimes does the new score add value?), but it does not yet answer the operational question of how the measures would have performed under sequential, realistic deployment. A natural extension is therefore a walk-forward backtest, in which the training window is rolled (or anchored and grown) day by day or month by month, the chosen formula is recomputed at each step, and positions are rebalanced into the following out-of-sample period. The cycle-level returns are then chained into a continuous equity curve, allowing path-dependent quantities such as cumulative profit and loss, realized drawdown, turnover, the Sharpe ratio of the strategy returns themselves, and the longest streak of losing cycles to be measured directly rather than inferred from per-configuration snapshots.

Such an evaluation would address three limitations of the present study. First, it would test whether the regime-dependent edge documented in Section 7 survives sequential rebalancing, in which the practitioner does not know the regime in advance and must commit capital before observing the next test window. Second, it would provide the natural setting for the regime-switching combination proposed in our Discussion: the realized-volatility, trend-strength, and dislocation-magnitude filters could be evaluated cycle by cycle, and the switched strategy compared against the pure momentum and pure reversion baselines on the same path. Third, it would expose the measures to realistic frictions (transaction costs, turnover penalties, and short-borrow constraints) that are immaterial in a one-shot test but accumulate decisively over many cycles. A walk-forward extension would therefore convert the retrospective regime taxonomy of this report into a forward-looking strategy evaluation, and is the most natural next step for the project.

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